

ECE 315: Signals and Systems

Homework 5

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This homework is due on **3/30, 11:59pm**. For analytical problems, write your solutions independently by hand. For coding problems, attach your source code and answers. Combine them in one PDF document and submit on CANVAS. Please ensure that your PDF is clearly readable on a computer.

1. (2pts: Periodic signals, Harmonics) Consider the following continuous-time signals:

- $2 \sin(t) + \cos(3t)$
- $\sin(3t) + 7 \sin(\pi t)$
- $(\sin(2t))^2$
- $(\sin(t) + \sin(3t))^2$

Determine whether they are periodic or aperiodic. For periodic signals, determine their harmonic components.

2. (2pts: Fourier series representation) Consider the following Fourier series coefficients for each continuous-time periodic signal $x(t)$:

- $a_k = (-1)^k \frac{\sin(k\pi/4)}{k\pi}$ for $k \neq 0$, $a_0 = \frac{1}{4}$
- $a_k = \begin{cases} 1 & \text{when } k = \text{even} \\ -1 & \text{when } k = \text{odd} \end{cases}$

Assume the signal $x(t)$ has period 4. Determine the signal $x(t)$ for each case.

3. (2pts: Fourier series coefficients) Consider the following continuous-time periodic signals:

- $x(t)$ is periodic with period 1 and $x(t) = e^{-t}$ for $0 < t < 1$;

- $x(t)$ is periodic with period 2 and $x(t) = \begin{cases} \sin(\pi t) & \text{when } 0 \leq t \leq 1 \\ 0 & \text{when } 1 \leq t < 2 \end{cases}$

Compute their Fourier series representations.

4. (2pt: Fourier series properties) Assume a continuous-time periodic signal $x(t)$ with fundamental period T has Fourier series coefficients a_k . We consider the following transformed signals:

- $x(t - t_1) + x(t + t_2)$
- $x(2t - 1)$
- $\text{Re}(x(t))$
- $\frac{d^2 x(t)}{dt^2}$

Compute the Fourier series coefficients for each signal.

5. (2pt: Fourier series properties) Consider the following three continuous-time signals with a period $T = 1/4$:

- $x(t) = \cos(8\pi t)$
- $y(t) = \sin(8\pi t)$
- $z(t) = x(t + 1)y(t - 1)$

- (i) Determine the Fourier series coefficients of $x(t)$, $y(t)$, and $z(t)$ using Euler's formula;
- (ii) Determine the Fourier series coefficients of $z(t)$ using the multiplication property of Fourier series coefficients. Check whether they are the same as those in (i).

6. Read Sections **6.1-1** and **6.7-1**, and complete the MATLAB problems **6.7-1** and **6.7-2** in **LSS**.